

Mrudula A P

Data Analyst || Data Scientist || AI/ML Engineer

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SKILLS

- **Languages:** Python, SQL
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn, TensorFlow, Keras, PyTorch, NLTK, SpaCy, OpenCV, Flask, Streamlit, MLFlow
- **Data Analysis & Visualization:** Tableau, Power BI, Data Mining, Feature Engineering, Data Wrangling, Time Series Forecasting, Regression Analysis, Hypothesis Testing, A/B Testing, Product Analytics, Business Intelligence (BI), KPI Reporting, Exploratory Data Analysis (EDA), Data Storytelling, Probability & Statistics, Statistical Analysis, Data Quality Assurance, Data Validation,
- **Machine Learning & Predictive Modelling:** Supervised & Unsupervised Learning, Recommender Systems, K-Means Clustering, Linear Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, Support Vector Machine (SVM), Ensemble Methods, Model Deployment, Model Interpretability (SHAP, LIME), Feature Selection, Hyperparameter Tuning, Cross-Validation, Dimensionality Reduction (PCA, t-SNE, UMAP), Model Monitoring, ML Lifecycle, MLOps, Reinforcement Learning
- **Deep Learning, Computer Vision & NLP:** Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Generative Adversarial Networks (GANs), Autoencoders, Transfer Learning, Transformers, Sequence-to-Sequence Models, Image Augmentation, Object Detection, Semantic Segmentation, YOLO, Image Classification, Computer Vision Pipelines, Model Optimization, BERT, Text Classification, Named Entity Recognition (NER), Sentiment Analysis, Topic Modelling (LDA), Text Summarization, Chatbot Development, Speech Recognition, TF-IDF, Word2Vec, Word Embeddings, Tokenisation
- **Generative AI, LLMs & Agentic AI:** GPT-4, BERT, Claude, Llama, Langchain, Lang Graph, Prompt Engineering, Retrieval-Augmented Generation (RAG), FAISS, Pinecone, Hugging Face, LoRa, QLoRA, PEFT, Semantic Search, Chatbot Development, Agentic AI, Autonomous Agents

EXPERIENCE

Data Associate L1

Apr 2025 - Present

Infotact Solutions Pvt Ltd.

- Contributed to the development and evaluation of machine learning models for predictive analytics, improving model accuracy and performance metrics.
- Designed and implemented an automated task management system using Python, improving task tracking efficiency by 40% through data-driven workflow optimization.

Data Science Intern

Nov 2023 - Sep 2024

Meta Scifor Technologies Pvt. Ltd.

Bangalore, Karnataka

- Spearheaded the integration of deep learning models for image recognition, achieving a 17% improvement in object detection accuracy.
- Established KPIs with stakeholders and developed monitoring dashboards, delivering actionable insights that improved model performance by 25%.

Assistant Technical Officer

Sep 2023 - Oct 2023

InterGlobe Enterprises (IndiGo)

Bangalore, Karnataka

- Collaborated with **engineering** and **maintenance teams** to **diagnose issues** and provide **technical solutions**, improving operational efficiency and reducing maintenance costs.
- Utilized data analysis tools to track maintenance performance, identifying trends and areas for process improvement.

Data Science & Machine Learning Intern

Feb 2023 - Jun 2023

Innomatics Research Labs

Hyderabad

- Automated data analysis pipelines for machine learning use cases, reducing manual processes and errors by 10%.
- Constructed comprehensive data dashboards for the effective and timely visualization of data, increasing work efficiency by 20%.

Assistant Professor

Feb 2019 - Jun 2019

NSS College of Engineering

Palakkad, Kerala

- Mentored and guided undergraduate students in the Department of Instrumentation & Control Engineering, enhancing their academic performance and securing a 40% increase in placements through targeted support and career advice.
- Served as a Committee Member for Question Paper Scrutiny Board at APJ Abdul Kalam Kerala Technical University.

Subject Matter Expert

May 2018 - Jan 2019

Orbit Institute of Science

- Revamped project-based learning activities, increasing student engagement by 25% and enhancing troubleshooting skills by 10%.
- Implemented creative teaching methods for Mathematics and Physics, integrating multimedia tools and practical experiments to elevate student understanding and interest.

EDUCATION

Scaler Academy

Certification in Data Science and Machine Learning with AI

2024

Amrita Vishwa Vidyapeetham

MTech in Control and Instrumentation Engineering | 9.12 CGPA

2017

NSS College of Engineering, Kerala

BTech in Instrumentation and Control Engineering | 7.84 CGPA

2015

ACADEMIC PROJECTS

Laptop Price Predictor Application (Flipkart)

<https://bit.ly/43mZTng>

- Developed a machine learning model for predicting laptop prices on Flipkart with 90% accuracy using regression algorithms, while collaborating on data cleaning, feature engineering, and exploratory data analysis.

Loan Eligibility Evaluator (LoanTap)

<https://bit.ly/46UhmES>

- Evaluated feasibility for extending tailored credit line loans to millennials using EDA and classification algorithms achieving an F1 score of 0.85 and a precision improvement of 10%, alongside Precision-Recall Trade-off analysis.

Demand Analysis of Shared Electric Cycles in the Indian Market (Yulu)

<https://bit.ly/3Sk3Cih>

- Developed a web application using Streamlit for users (<https://bit.ly/3Og9HJw>) to know shared bike demand nearby.
- Conducted in-depth analysis using Pandas, NumPy, Seaborn, and Matplotlib. Identified key predictors with hypothesis testing techniques. Used 2-sample t-test, ANOVA, and Chi-Square with SciPy.

Temporal Analysis for Enhancing Digital Advertising Efficiency (AdEase)

<https://bit.ly/3SpHB1A>

- Analyzed 550 days of page view data using Time Series forecasting techniques, generating accurate forecasts that optimized targeted advertising strategies, improving **ad campaign effectiveness by 20%** and increasing **client engagement by 15%** across diverse global markets.

Personalization systems for an Indian Entertainment Conglomerate (Zee)

<https://bit.ly/3FISPGI>

- Implemented a recommender model achieving 85% precision in delivering personalized movie recommendations, utilizing Collaborative Filtering, Pearson Correlation, Cosine Similarity, and Matrix Factorization techniques.

Optimizing Order Placement and Fulfillment for Global Retailer

<https://bit.ly/3MqtqWi>

- Analyzed 100,000 orders from 2016-2018 using SQL, extracting actionable insights that led to a 15% increase in sales through targeted recommendations and strategic adjustments.

Open Pub Application

<https://bit.ly/3lJYzkZ>

- Developed a multi-page web application using Streamlit to help users find nearby pubs with **92%** accuracy.
- Enabled searches by Postal Code or Local Authority, displaying the nearest 5 pubs based on Latitude and Longitude.

Revolutionizing Tweet Analysis with Advanced NER Technology

<http://bit.ly/3TINXcD>

- Engineered a Named Entity Recognition (NER) system for Twitter data using Python, TensorFlow, and Keras, enhancing accuracy by 25% with LSTM + CRF, and achieving 95% accuracy using BERT-base-uncased from Hugging Face Transformers.

Vegetable Classification for Ninjacart's Supply Chain Automation

<https://bit.ly/3UVvrBc>

- Spearheaded VegiVision for Ninjacart, achieving 95% accuracy in classifying onions, potatoes, and tomatoes using image classification, improving model precision through transfer learning and reducing overfitting by 20%.